Interconnection Networks

CSE Sem-4

Def. Interconnection network are the network infrastructure for parallel processing. Network connect processors to memory with optional switch element.

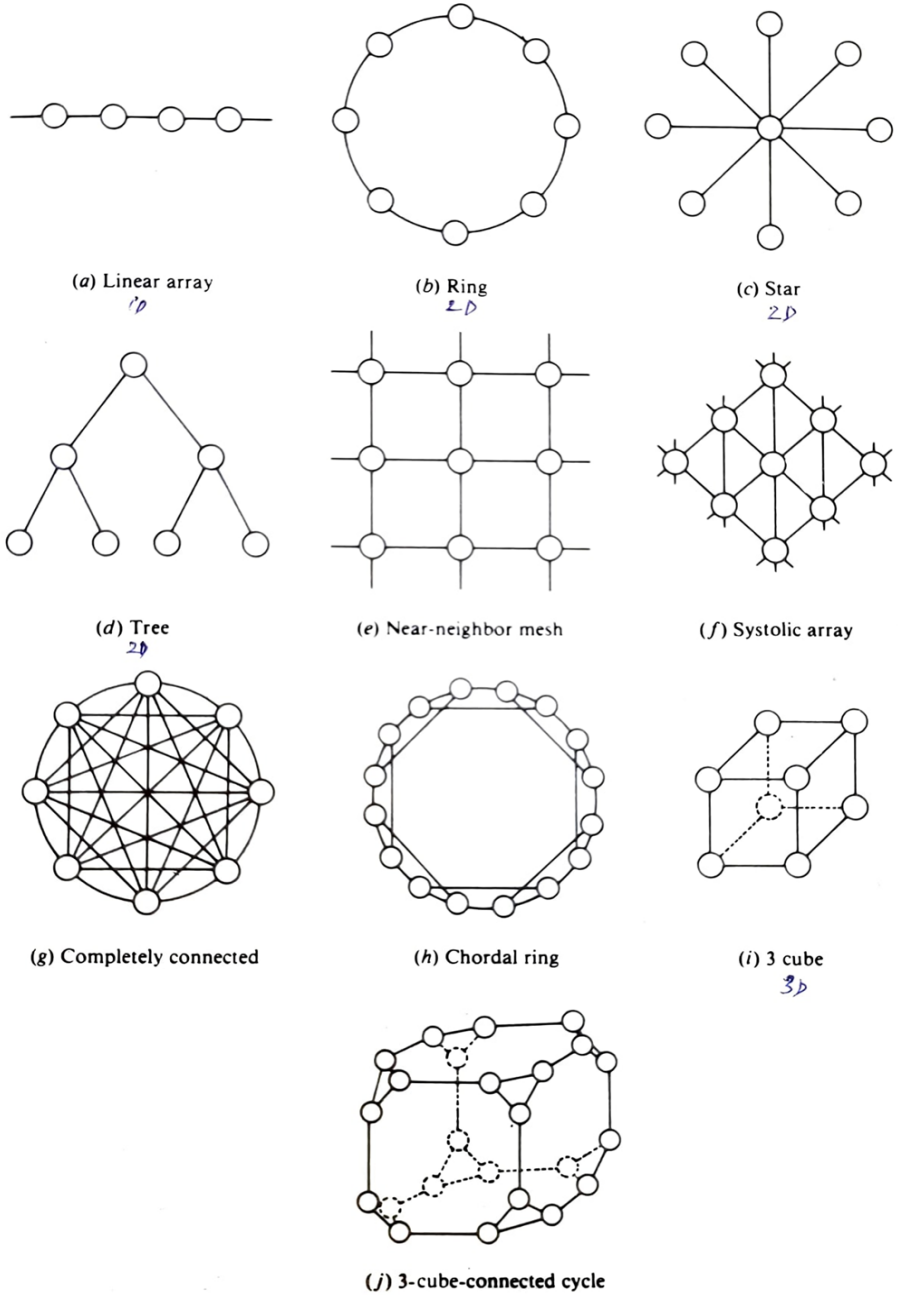
- Interconnection networks can transfer data between processors and memory. These are made of switches and links (wires, fiber).
- Classification of network topology: **Static Networks**, **Dynamic Networks**

Static Networks

- **Static Networks:** These networks are formed by point-to-point direct connections, which will not change during program execution.

- Types of Static Networks:

1. 1-D : Linear array (used for pipeline architecture, N nodes relate to N-1 links)
2. 2-D : Ring (Two terminal nodes of a linear array connected with one extra link), star, tree, mesh and systolic array
3. 3-D : Completely connected ring, 3 cube (degree of each node is 3), 3-cube-connected-cycle (CCC)



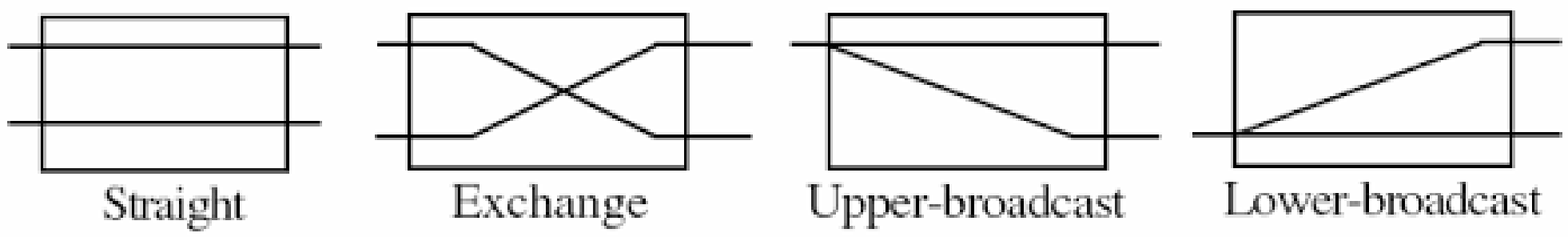
Dynamic Networks

- **Dynamic networks:** These are implemented with switches, which are dynamically configured to match the communication demand in user programs.
- **Types of Dynamic Networks**: Single stage network, multistage network, crossbar network
- **Single stage network:**

- A single stage of switching elements (SEs) exists between the i/ps and the o/ps of the network.
- The simplest SE that can be used is the 2*2 SE (which is 2*2 switch).
- This switch has 2 i/ps and 2 o/ps. Messages arriving on either i/p line can be switched to either o/p line.

The connection in a 2×2 switch can be of four types:

- In the **straight** setting, the upper i/p is transferred to the upper o/p and the lower i/p is transferred to the lower o/p.
- In the **exchange** setting, the upper i/p is transferred to the lower o/p and the lower i/p is transferred to the upper o/p.
- In the **upper-broadcast** setting the upper i/p is broadcasted to both the upper and the lower outputs.
- In the **lower-broadcast** setting the lower i/p is broadcasted to both the upper and the lower outputs



Multistage Interconnection Network (MIN)

- **Multistage Interconnection Networks (MINs)** are a class of high-speed computer networks where the first stage of links is connected to the sources (usually processors) and the last stage is connected to the destinations (memory modules).

- MINs have been classified into **two classes** depending on the availability of paths to establish new connections:

1. **Blocking:** A connection between input/output pair is not always possible because of conflicts with existing connections. Typically, there is a unique path between every input/output pair. A uni-path network is also called a **Banyan Network**.

A Banyan network is defined as a class of multistage interconnection networks in which there is one and only one path from any input node to any output node.

Delta Networks, are a subset of Banyan networks. The routing property of Delta Network is called a **Self routing Property** or **Delta Property**.

- Types of Delta Networks: **Omega Network, Butterfly Network**

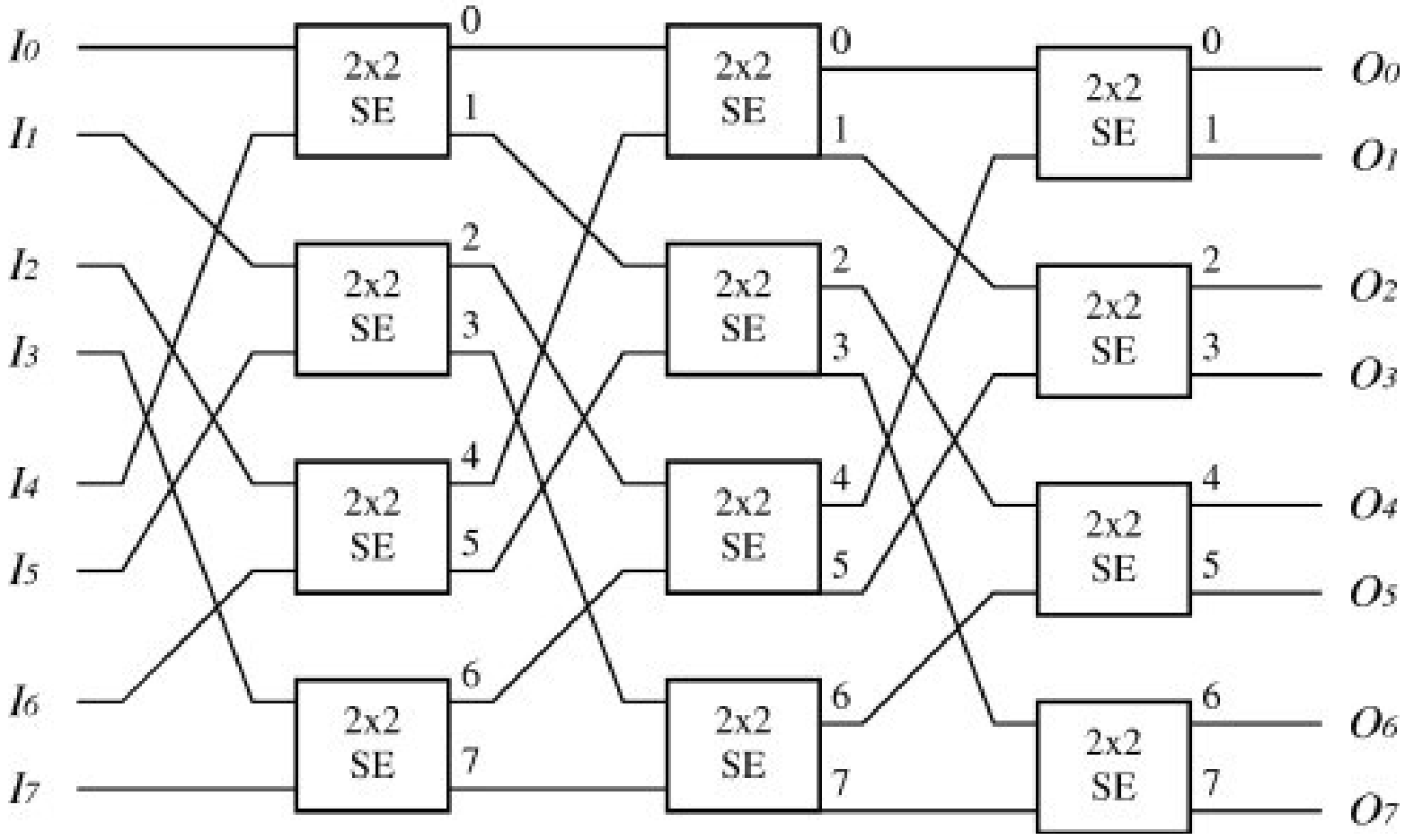
2. **Non Blocking:** Any input can be connected to any free output port without affecting the existing connections. They require extra stages and have multiple paths between every input and output. A popular example of Non-blocking networks is a Clos network.

Omega Network

❖Omega networks are the most popular of Delta networks.

• The interconnections between stages are defined by the **logical rotate left** of the bits used in the port ids.

000 → 000 → 000 → 000
001 → 010 → 100 → 001
010 → 100 → 001 → 010
011 → 110 → 101 → 011
100 → 001 → 010 → 100
101 → 011 → 110 → 101
110 → 101 → 011 → 110
111 → 111 → 111 → 111

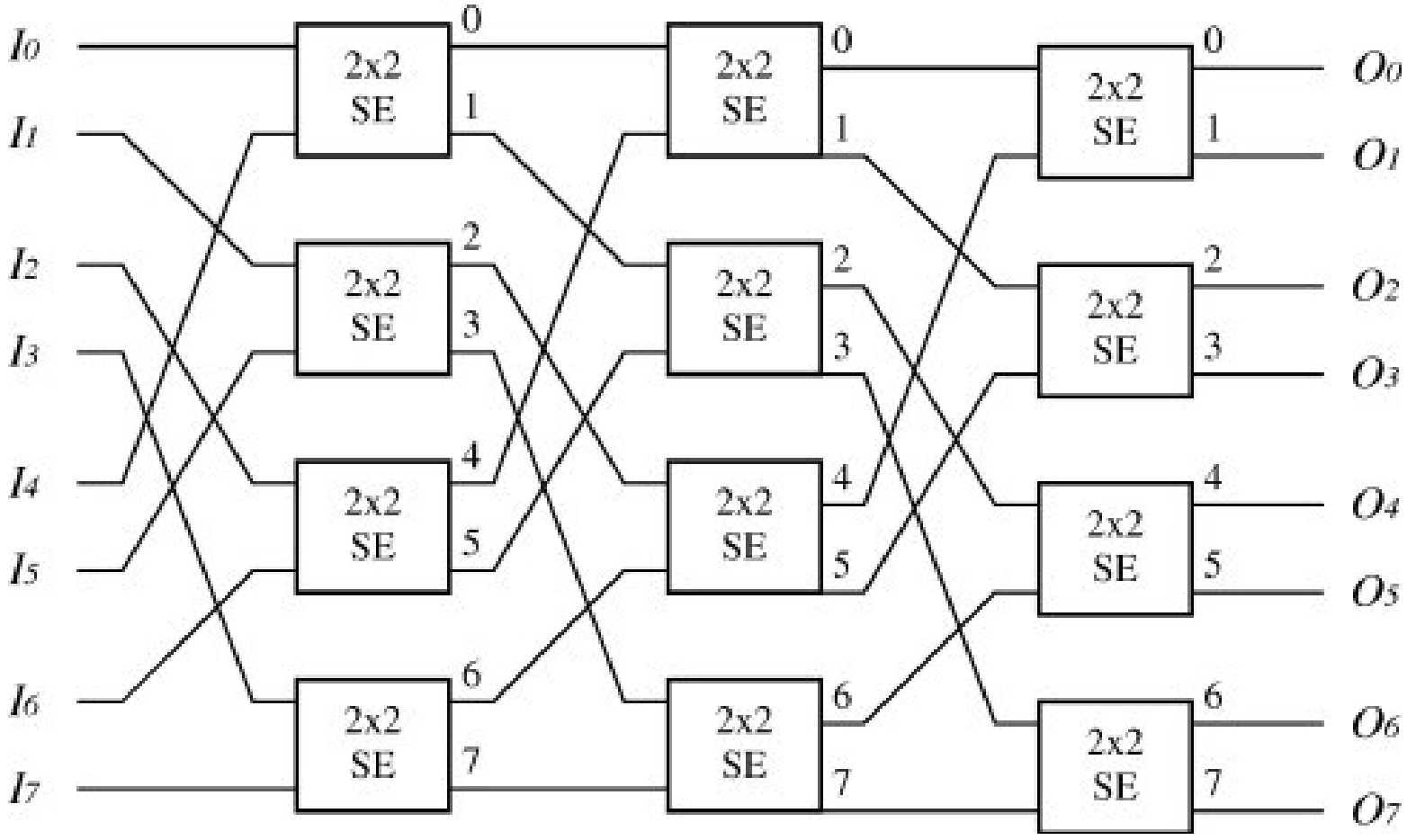


Omega Network

For (NxN) omega network,

- No of stages: $\log_2 N$
- No of switches in each stage: N/2

- Each bit in the destination address can be used to route the message through one stage.
- The destination address bits are scanned from left to right and the stages are traversed from left to right.
- The first MSB is used to control the routing in the first stage; the next bit is used to control the routing in the next stage, and so on.
- If the bit in the destination address is 0, the message is routed to the upper output of the switch. If the bit is 1, the message is routed to the lower output of the switch.



Crossbar Switch Networks

- **Crossbar Switch** system contains of a few cross points that are kept at intersections among memory module and processor buses paths.
- In each cross point, the small square represents a switch Each switch point has control logic to set up the transfer path among a memory and processor.
- It calculates the address which is placed in the bus to obtain whether its specific module is being addressed.
- In addition, it eliminates multiple requests for access to the same memory module on a predetermined priority basis.

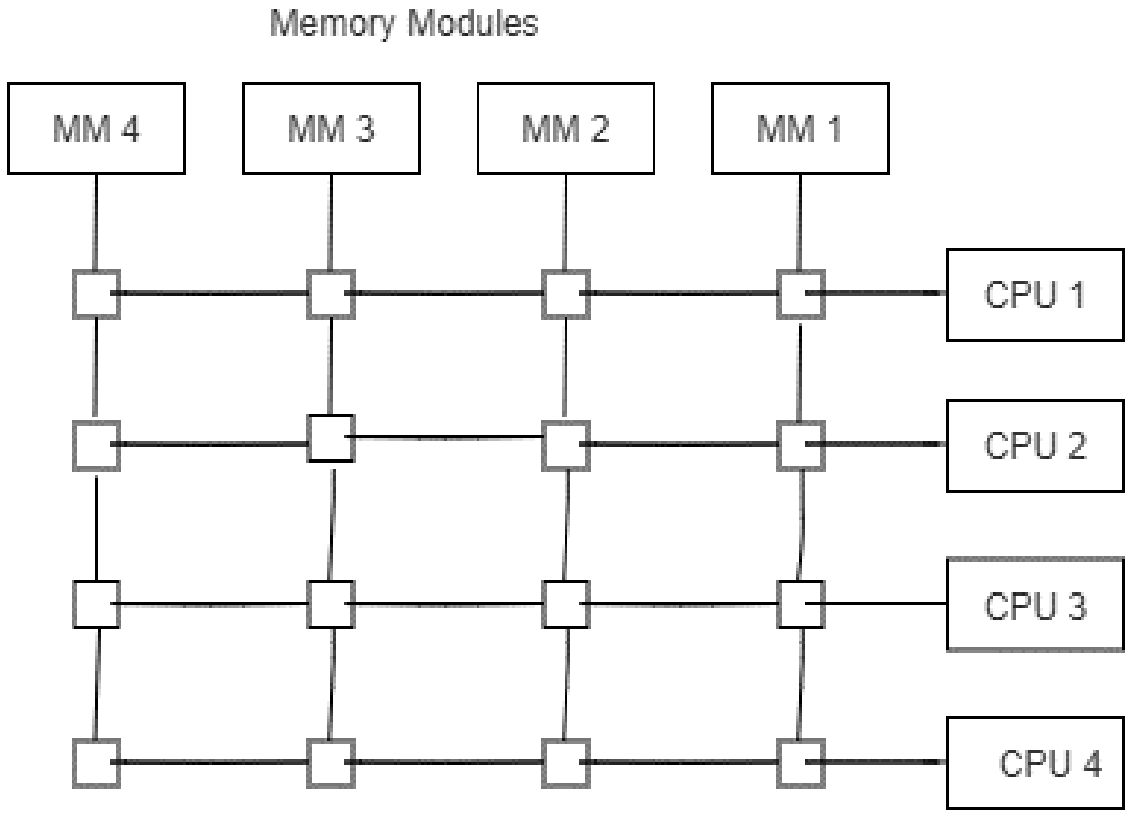


Figure - Crossbar Switch System